

MultiTech Escape PUZZLE GAME - SETUP GUIDE

This guide will help you set up and start building your arrow puzzle game.

1. INSTALL DOTWEEN (REQUIRED!)

The game uses DOTween for smooth animations. You must install it:

METHOD 1: Unity Asset Store (Recommended - FREE)

1. Open Unity Asset Store (Window > Asset Store)
2. Search for "DOTween (HOTween v2)"
3. Download and Import (it's free!)
4. Or visit: <https://assetstore.unity.com/packages/tools/animation/dotween-hotween-v2-27676>

METHOD 2: Package Manager (DOTween Pro - Paid)

1. If you have DOTween Pro, add via Package Manager
2. Window > Package Manager > Add package from git URL
3. Enter your DOTween Pro package URL

After installation, you may need to setup DOTween:

- A setup panel may appear automatically
- If not, go to Tools > Demigiant > DOTween Utility Panel
- Click "Setup DOTween"

2. CREATE ARROW SPRITES

You need to create arrow sprites for your game:

Required Sprites:

- Arrow sprite (pointing RIGHT) - This will be rotated for other directions
- Recommended size: 64x64 pixels or 128x128 pixels
- Format: PNG with transparency
- Style: Match the screenshots (thick maze-like arrow)

Where to place sprites: Assets/Sprites/Arrows/

You can:

1. Create sprites in any image editor (Photoshop, GIMP, Aseprite, etc.)
2. Use the built-in Unity sprites (Square/Circle)
3. Find free arrow sprites online

For a quick test, you can use Unity's built-in sprites:

- Create a UI Image
- Use Unity's default "Knob" or "UISprite" as placeholder

IMPORTANT - Sprite Import Settings:

If a sprite appears too small or too big when applied via the Theme Editor window:

1. Select the sprite in the Project window
 2. In the Inspector, find "Pixels Per Unit" in Import Settings
 3. Adjust the value (typically between 100-1000 depending on sprite resolution)
 4. Click "Apply" to save changes
 5. Re-apply the theme in the Theme Editor
6. CREATE ARROW PREFAB
7. Create Empty GameObject: "Arrow"
 8. Add Components:
 - SpriteRenderer (assign your arrow sprite)
 - BoxCollider2D (for tap detection)
 - ArrowPiece script
 - (Optional) ArrowVisuals script
 3. Configure:
 - Set sprite color to white
 - Set collision color to red

- Adjust collider size to match sprite

4. Save as Prefab:

- Drag to Assets/Prefabs/Arrows/
- Name it "ArrowPrefab"

4. SETUP GAME SCENE

Create the main game scene:

1. CREATE GRID OBJECT:

- Create Empty GameObject: "Grid"
- Add Unity Grid component (Create 2D Object > Grid)
- Add Tilemap component (will appear automatically)
- Add Tilemap Renderer
- Add GridBoard script

2. CREATE GAME MANAGER:

- Create Empty GameObject: "GameManager"
- Add GameManager script
- Add LevelManager script
- (Optional) Add ThemeManager script for runtime theme switching

3. CREATE INPUT HANDLER:

- Create Empty GameObject: "InputHandler"
- Add InputHandler script
- Assign Main Camera

4. CREATE UI:

- Create Canvas (UI > Canvas)
- Set to Screen Space - Overlay
- Create UI elements:

- Level number text (top center)
- Hearts container (top center)
- Add UIManager script to Canvas
- Assign references

5. CONFIGURE CAMERA:

- Set to Orthographic
- Adjust size to fit grid (try 6-8)
- Position: (6, 6, -10) for a 12x12 grid

5. CREATE YOUR FIRST LEVEL

Using the Arrow Paint Tool:

1. Open scene with GridBoard
2. Select the Grid GameObject
3. Go to Tools > Arrow Puzzle > Activate Arrow Paint Tool
4. OR press 'T' and select "Arrow Paint Tool" from toolbar

Controls:

- Press 1: Set direction to UP
 - Press 2: Set direction to RIGHT
 - Press 3: Set direction to DOWN
 - Press 4: Set direction to LEFT
 - Click/Drag: Paint arrows
 - Alt+Click: Delete arrow
 - Ctrl/Cmd+S: Save level data
6. Save level as ScriptableObject to Assets/Levels/
 7. CONFIGURE LEVEL MANAGER
 8. Select GameManager GameObject

9. Find LevelManager component
10. Assign Arrow Prefab
11. Add your created level to "All Levels" list
12. (Optional) Set as "Current Level Data" for testing
13. (Optional) In "Theme (Optional)" section, assign CurrentTheme asset for runtime theme support

6A. SETUP RUNTIME THEME SYSTEM (OPTIONAL)

The theme system supports runtime theme switching. To enable it:

1. APPLY A THEME FIRST:

- Open: Tools > MultiTech Studio > Escape Game Setup
- Go to Theme tab
- Select a theme and click "APPLY"
- This automatically creates CurrentTheme.asset in Themes folder

2. ADD THEME MANAGER:

- Create Empty GameObject: "ThemeManager"
- Add ThemeManager component
- Assign CurrentTheme asset (found in Assets/MultiTech Studio/Escape Game/Themes/)
- Assign scene references:
 - LevelManager: Drag your LevelManager GameObject
 - GridManager: Drag your GridManager GameObject
 - Background Renderer: Drag the Background GameObject's SpriteRenderer
 - Theme Icon Image: Drag the Icon UI Image component (if exists)
- Enable "Apply On Start" to auto-apply theme when scene loads

3. (OPTIONAL) CONFIGURE LEVEL MANAGER:

- Select GameManager GameObject
- In LevelManager component, find "Theme (Optional)" section
- Assign the same CurrentTheme asset

- This allows LevelManager to use theme's arrow prefab at runtime

4. RUNTIME THEME SWITCHING:

- Access ThemeManager via: ThemeManager.Instance
- Change theme: `currentThemeAsset.ActiveTheme = newTheme`
- ThemeManager will automatically apply all theme settings

NOTE: The runtime theme system is optional. The game works fine without it, using the traditional editor-based theme application method.

7. PROJECT SETTINGS FOR MOBILE

For mobile deployment:

1. Build Settings:

- File > Build Settings
- Switch to iOS or Android

2. Player Settings:

- Resolution: Portrait or Landscape
- Supported Orientations: Choose one
- Minimum API Level: Android 7.0+
- Target iOS: 13.0+

3. Input Settings:

- InputHandler is already configured for touch
- Uses Unity's new Input System

8. RECOMMENDED LAYER SETUP

Create layers for better organization:

- Arrows (Layer 6)
- Grid (Layer 7)
- UI (Layer 5 - default)

Assign layers to objects:

- Arrow prefab > Layer: Arrows
- Grid > Layer: Grid
- Canvas > Layer: UI

Update InputHandler:

- Set "Arrow Layer Mask" to "Arrows" layer only

9. TESTING

Test your game:

1. Enter Play Mode
2. Click/tap on arrows to move them
3. Arrows should:

- Move in their direction
- Exit if path is clear
- Turn red and return if blocked

If arrows don't respond:

- Check arrow has BoxCollider2D
- Check InputHandler has correct camera reference
- Check arrow sprite is visible

10. SDK INTEGRATION (AdMob & Firebase)

The game supports optional integration with Google Mobile Ads (AdMob) and Firebase.

SETTING UP SDKs:

1. Install SDKs (if needed):

- AdMob: Import Google Mobile Ads SDK from Package Manager or Asset Store
- Firebase: Import Firebase SDK from Package Manager or Asset Store

2. Access SDK Settings:

- Open: Tools > MultiTech Studio > Escape Game Setup

- Click on the "Settings" tab

3. SDK Status:

- The window shows current SDK availability
- "✓ Available" means the SDK is detected in your project
- "X Not Available" means the SDK is not installed

4. Enable/Disable SDKs:

- Check/uncheck "Enable Google Mobile Ads (AdMob)" to toggle AdMob
- Check/uncheck "Enable Firebase" to toggle Firebase
- The system uses conditional compilation - code compiles safely with or without SDKs

5. Configure AdMob (when enabled):

- Add AdsManager component to a GameObject in your scene
- In the Settings tab, configure:
 - Banner ID
 - Interstitial ID
 - Rewarded Interstitial ID
 - Rewarded ID
 - Interstitial Level Threshold (minimum level to show ads)
 - Can Show Interstitial After Level (enable/disable interstitial ads)

6. Firebase Setup:

- Add FirebaseManager component to a GameObject in your scene
- Configure Firebase in your Firebase console
- The system automatically initializes Firebase when enabled

7. Actions:

- "Update Defines": Manually refresh SDK detection
- "Clear All SDK Defines": Remove all SDK defines for testing
- "Test Compilation": Verify project compiles correctly

- "Refresh Project": Refresh Unity's asset database

IMPORTANT NOTES:

- SDKs are optional - the game works without them
- Conditional compilation ensures code compiles safely
- AutoDefines script automatically detects SDKs on project load
- You can enable/disable SDKs at any time without breaking the project

11. NEXT STEPS

Enhance your game:

1. Create more levels
2. Add sound effects
3. Add particle effects for exits/collisions
4. Create main menu scene
5. Add level select screen
6. Implement scoring system
7. Add undo functionality
8. Create tutorial levels
9. Integrate AdMob for monetization
10. Set up Firebase for analytics

TROUBLESHOOTING

"DOTween namespace not found": → Install DOTween from Asset Store

"Arrow doesn't respond to clicks": → Check BoxCollider2D is attached → Check InputHandler has camera assigned

"Arrow sprite not showing": → Check SpriteRenderer has sprite assigned → Check camera is orthographic and positioned correctly

"Grid not visible in editor": → Enable Gizmos in Scene view → Check GridBoard "Show Debug Gizmos" is checked

"Sprite appears too small or too big after applying theme": → Adjust the sprite's "Pixels Per Unit" setting in Import Settings → Select the sprite in Project window, then in Inspector adjust "Pixels Per Unit" → Typical range: 100-1000 depending on sprite resolution → Click "Apply" and re-apply the theme

"AdsManager not found in scene": → Add AdsManager component to a GameObject in your scene → The component must be in the active scene for configuration

"SDK not detected": → Ensure the SDK is properly imported into your project → Check that SDK assemblies are referenced → Click "Update Defines" in Settings tab to refresh detection → Verify SDK installation in Package Manager or Asset Store

"Compilation errors with SDK enabled": → Ensure SDK is properly installed and imported → Check SDK version compatibility with your Unity version → Try "Test Compilation" button in Settings tab → If issues persist, disable the SDK temporarily using the toggle

"ThemeManager not applying theme at runtime": → Ensure CurrentTheme asset is assigned to ThemeManager → Check that all scene references are assigned (LevelManager, GridManager, etc.) → Verify "Apply On Start" is enabled if you want auto-apply on scene load → Check Unity console for error messages

"CurrentTheme asset not found": → Apply a theme in the Theme Editor (Tools > MultiTech Studio > Escape Game Setup) → This automatically creates CurrentTheme.asset in Themes folder → Or manually create: Right-click > Create > Tools/MultiTech Studio/Escape Game Current Theme

SUPPORT

For issues or questions:

1. Check scripts for TODO comments
2. Review Unity console for errors
3. Ensure all dependencies are installed
4. Verify all references are assigned in Inspector

Happy game making!